Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

AUGUST/RESIT EXAMINATIONS 2022/2023

MODULE: CA119 - Data Science and Databases

PROGRAMME(S):

DS	BSc in Data Science
AP	BSc in Applied Physics
ECSAO	Study Abroad (Engineering & Computing)

YEAR OF STUDY: 1,3,O

EXAMINER(S):

Mark Roantree	(Internal)	(Ext:5636)
Assoc. Prof. Sheila McBreen	(External)	External
Prof. Gerard O'Connor	(External)	External
Dr. Ziqi Zhang	(External)	External

TIME ALLOWED: 2 Hours

INSTRUCTIONS: **Answer 4 questions. All questions carry equal marks.**

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1 **[TOTAL MARKS: 25]**
Entity Relationship Modelling

Q 1(a) **[12 Marks]**
A plan to design a university database must capture *Students*, *Professors*, *Courses*, *Modules* and *ClubAndSocs*. A Course has multiple modules but a module can be taught on one course only. Professors teach modules. Students can register for 1 Course but join any number of Clubs or Societies.

Provide a separate Entity-Relationship diagram for each entity. Ensure that you include at least 1 derived attribute; 2 multi-valued attributes; 2 composite attributes; and 1 complex attribute.

Underneath each E-R diagram, write the names and types of the 4 different attributes. Ensure that they are properly represented in the E-R diagram.

Q 1(b) **[6 Marks]**
Provide one example of each of the following relationships: one-to-one, one-to-many, and many-to-many. Draw a new E-R diagram to represent these relationships. You may draw a single diagram or more than one.

Q 1(c) **[7 Marks]**
In relationships, what is meant by *participation constraint*?

Provide an example of each constraint type from your relationship examples in part (b) of this question. Explain why your two chosen examples have these constraints.

[End of Question1]

QUESTION 2 **[TOTAL MARKS: 25]**
Relational Algebra

This question uses Appendix A.
Assume we denote the relations WorldConference as *W* and Temperatures as *T*.

Q 2(a) **[8 Marks]**
Why are *W* and *T* not union compatible?
Write the algebraic expression(s) that will provide union compatibility for *W* and *T*.
Write the algebraic expression for “For what cities do we have *both* conference venue and temperature?” Show the result table for this expression.

Q 2(b) **[12 Marks]**

Show the results (relations) for each of the following expressions.

(i) W ⋈ T

(ii) W ⟕ T

(iii) W ⟖ T

Q 2(c) **[5 Marks]**

Using *W*, demonstrate that commutativity does not hold using the PROJECT operation.

[End of Question 2]

QUESTION 3 **[TOTAL MARKS: 25]**

Normalisation

This question uses Appendix B.

A Company uses an Invoice which must be captured into a database as shown in Appendix B. Here, we can see the first normal form version of the schema.

ORDER [<u>Order_no</u> ,Order_date, Cust_name, Cust_addr, Ship_addr, Phone1, Phone2, Cust_disc (item#, Prod_code, Prod_desc, Qty, Filled, Price)]

Q 3(a) **[6 Marks]**

What is meant by First Normal Form (1NF)?
What is the process for transforming an unnormalised table to a 1NF table?
Show the resulting changes to the unnormalised schema shown above.

Q 3(b) **[9 Marks]**

Define Second Normal Form (2NF).
What is required to transform a 1NF schema to 2NF?
Show the process for transforming your current 1NF schema into a 2NF schema.
(Hint: just one new table is required)

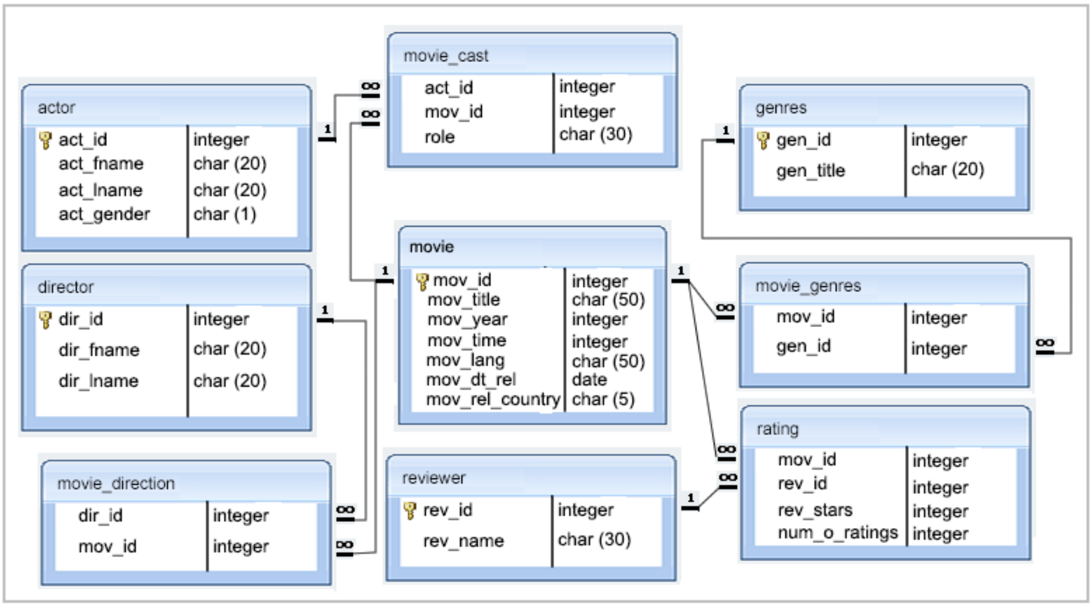
Q 3(c) **[10 Marks]**

Define Third Normal Form (3NF).
What is required to transform a 2NF schema to 3NF?
Show the process for transforming your current 2NF schema to 3NF?
(Hint: just one new table is required)

[End of Question 3]

QUESTION 4 **[TOTAL MARKS: 25]**

SQL Programming



Study the above schema which describes a movie database. It displays table names, the attributes contained within each table, and the 1-to-many relationships that connect tables. Use SQL to answer the following queries.

Q 4(a) **[3 Marks]**

List all movies that were released in a country other than France.

Q 4(b) **[4 Marks]**

Find the list of all those movies and their year, which include the words “once upon” anywhere in the title.

Q 4(c) **[5 Marks]**

Find the titles of all movies that have no ratings.

Q 4(d) **[6 Marks]**

Find the reviewer's name and the title of the movie for those reviewers who rated more than one movie.

Q4 (e) **[7 Marks]**

Find those movies with the shortest running time, along with the year, director name, actor name and his/her role in that production.

[End of Question 4]

QUESTION 5 **[TOTAL MARKS: 25]**

Relational Model

This question uses Appendix A.

Assume we denote the relations WorldConference as *W* and Temperatures as *T*.

Q 5(a) **[4 Marks]**

What is the Degree of W?
What is the cardinality of W?
What is the degree of the Cartesian product of W and T?

Q 5(b) **[8 Marks]**

What is the key for T?
Write down all superkeys of T.
What is the key for W?
Write down all superkeys of W.

Q 5(c) **[8 Marks]**

Using relation T, provide an example of 4 possible constraint violations during an INSERT operation.

Q 5(d) **[5 Marks]**

What is meant by the term *Referential Integrity*?
Assume we had a third table called **Cities**. How might the referential integrity constraint be defined between **Cities** and an updated version of **Temperatures**?

[End of Question 5]

Appendix A

WorldConference

Date	Location	Attendance
Sep-06	New York	1100
Mar-07	New York	1050
Sep-07	San Francisco	2000
Mar-08	New York	1200
Sep-08	San Francisco	1850
Mar-09	Boston	1400
Sep-09	Boston	1450
Mar-10	New York	1200
Sep-10	New York	1100
Mar-11	Boston	1500
Sep-11	New York	1200

Temperatures

Date	City	AvgTemp
Mar-07	San Francisco	22
Mar-07	New York	14
Sep-07	San Francisco	26
Sep-07	New York	27
Mar-08	San Francisco	21
Mar-08	New York	12
Sep-08	San Francisco	28
Sep-08	New York	26
Mar-09	San Francisco	19
Mar-09	New York	11
Sep-09	San Francisco	25
Sep-09	New York	25
Mar-10	San Francisco	24
Mar-10	New York	23

WorldConference is a list of venues for the world database conference together with their participation.
Dom(Date) is a string formatted as CCC-NN
Dom(Location) is a string
Dom(Attendance) is an integer with values in the range 300 to 3000.

Temperatures is a list of cities with their average monthly temperature.
Dom(Date) is a string formatted as CCC-NN
Dom(City) is a string
Dom(AvgTemp) is an integer with values in the range -20 to 100.

Appendix B

ABC MANUFACTURING ORDER FORM DATE: AUG 30, 2002

ORDER # 9932
MR. S.D. KURTZ
123 THAT STREET
TORONTO, ONTARIO
A9B 8C7

SHIPPING ADDRESS: 456 NO STREET
HAMILTON, ONTARIO
L6K 5J4

PHONE: (416) 879-0045 (416) 786-3241 CUSTOMER DISCOUNT: 3%

ITEM #	PRODUCT CODE	DESCRIPTION	QTY	BACKORDERED	FILLED	PRICE/UNIT	AMOUNT
1	FR223	HALF SIZE REFRIGERATOR	2	0	2	750.99	1501.98
2	TB101	PATIO TABLE	5	2	3	150.00	450.00
3	CH089	PATIO CHAIRS	20	0	20	35.00	700.00

TOTAL 2651.98
DISCOUNT AMT 79.56

AMOUNT OWING 2572.42

[END OF EXAM]